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PROJECT REPORT:

EXPLORATORY DATA ANALYSIS (EDA) ON ONLINE RETAIL DATASET USING SQL

Project Title: Uncovering Customer Behaviour and Sales Patterns: An Exploratory Data Analysis of an Online Retail Dataset Using SQL

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An undertaking is never a result of a solitary individual; rather it bears the engravings of various individuals who specifically or by implication helped in finishing that venture. We would bomb in my obligations on the off chance that we don't let out the slightest peep of gratitude to every one of the individuals who helped us in finishing this task of our own.

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PREFACE

This report presents a comprehensive Exploratory Data Analysis (EDA) of the "Online Retail" dataset, a transnational data set containing all transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based online retailer. The primary objective is to derive meaningful insights into customer purchasing behavior, sales trends, product performance, and key business metrics. The analysis is conducted in two stages: first using Python within a Google Colab environment for robust data cleaning, feature engineering, and visualization, and second using PostgreSQL to demonstrate how these insights can be operationalized within a structured database environment. The project leverages techniques such as univariate and bivariate analysis, time series analysis, cohort analysis, and RFM segmentation to create a holistic view of the business. Key findings include the identification of the United Kingdom as the dominant market, peak sales during weekdays and specific hours, a high concentration of revenue among a small number of customers, and distinct customer segments that can inform targeted marketing strategies.

This document serves as the final project report for the Internal Component-II of the Data Science-I course. It details the step-by-step process of conducting an Exploratory Data Analysis (EDA) on a real-world online retail dataset. The work is structured to follow the scientific method: starting with a clear objective, gathering data, cleaning it, performing analysis, and finally interpreting the results. The report includes both the theoretical background of the techniques used (like RFM and Cohort Analysis) and the practical implementation through code in Python and SQL. It is intended to demonstrate proficiency in data handling, statistical analysis, and deriving actionable business insights from raw data.

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1. INTRODUCTION

Exploratory Data Analysis (EDA) is a critical first step in any data science project. While Python offers rich visualization capabilities, SQL (Structured Query Language) remains the standard for interacting with relational databases. This report presents a SQL-centric analysis of the **Online Retail Dataset**, a classic transactional dataset from a UK-based gift retailer. The goal is to demonstrate how a data analyst can extract actionable business insights directly from a database using SQL, without relying on external programming languages. The analysis is conducted in **PostgreSQL** using **pgAdmin4**. We replicate many of the insights from the Python EDA but also leverage advanced SQL features such as window functions, common table expressions (CTEs), and conditional aggregation to perform customer segmentation, cohort retention, and revenue concentration calculations.

The dataset spans December 2010 to December 2011 and contains 541,909 records. After cleaning, two tables are used: retail_full (519,589 rows, includes transactions with missing Customer IDs for sales analysis) and retail_customers (387,844 rows, only transactions with valid Customer IDs for customer-centric analysis).

2. DATA SOURCE AND DESCRIPTION

Source: Kaggle – “Online Retail Dataset” by Vijaykumar Ummadisetty.

Citation: <https://www.kaggle.com/datasets/vijayuv/onlineretail?resource=download>

Original Dataset: 541,909 rows and 8 columns covering:

- **InvoiceNo** – invoice number (cancellations start with 'C')
- **StockCode** – product code
- **Description** – product name
- **Quantity** – number of units
- **InvoiceDate** – date and time of transaction
- **UnitPrice** – price per unit (£)
- **CustomerID** – unique customer identifier (many missing)
- **Country** – customer country

Cleaned Tables (Imported into PostgreSQL as CSV which were downloaded from the Google Colab after the cleaning and analysis were conducted in it):

- retail_full – all cleaned transactions (missing CustomerIDs retained for sales analysis)
- retail_customers – only rows with valid CustomerID (used for customer-centric analysis)

The tables include engineered features: Revenue, Year, Month, Month_Name, Day, Hour, DayOfWeek, InvoiceMonth, CohortMonth, CohortIndex.

3. DATABASE SCHEMA OVERVIEW

3.1 Table: retail_full

Column	Data Type	Description
Invoice_No	VARCHAR(20)	Transaction identifier
Stock_Code	VARCHAR(20)	Product identifier
Description	TEXT	Product description
Quantity	INT	Number of units
Invoice_Date	TEXT	Transaction date/time (original string)
Unit_Price	NUMERIC	Price per unit (£)
Customer_ID	VARCHAR(20)	Customer identifier (may be missing)
Country	VARCHAR(50)	Customer country
Revenue	NUMERIC	Quantity × Unit_Price
Year	INT	Year of transaction
Month	TEXT	Year-month in 'YYYY-MM' format
Month_Name	VARCHAR(20)	Full month name
Day	INT	Day of the month
Hour	INT	Hour of the day (0-23)
DayOfWeek	VARCHAR(20)	Day name (Monday-Sunday)

3.2 Table: retail_customers (Includes all columns from retail_full plus the following)

Column	Data Type	Description
InvoiceMonth	VARCHAR(10)	Month of transaction in 'YYYY-MM' format
CohortMonth	VARCHAR(10)	Customer's first purchase month
CohortIndex	INT	Months since first purchase (0,1,2,...)

4. SQL ANALYSIS METHODOLOGY

The SQL analysis was performed in **PostgreSQL** using **pgAdmin4**. The workflow consisted of:

1. **Table Creation** – Defining the schemas exactly as exported from Python.
2. **Data Import** – Loading CSV files (exported from Python) into the tables.
3. **Data Validation** – Verifying record counts, missing values, and data integrity.
4. **Exploratory Queries** – Replicating key Python analyses in SQL (basic metrics, time series, product/country performance).
5. **Advanced Analytics** – Implementing RFM segmentation, cohort retention, and revenue concentration using window functions and CTEs.
6. **Interpretation** – deriving business insights from the query outputs.

5. DATA VALIDATION AND CLEANING IN SQL

Before analysis, SQL queries were used to confirm that the imported data matched the cleaned Python datasets.

5.1 Record Counts

```
sql
SELECT COUNT(*) FROM retail_full;      -- 519,589
SELECT COUNT(*) FROM retail_customers; -- 387,844
```

Interpretation: The counts match the Python outputs, confirming successful import. The difference of 131,745 rows corresponds to records with missing CustomerIDs (24.9% of original).

5.2 Missing Values Check

```
sql
SELECT
  COUNT(*) FILTER (WHERE Description IS NULL) AS missing_description,
  COUNT(*) FILTER (WHERE Customer_ID IS NULL) AS missing_customer
FROM retail_full;
```

Result: 0 for both columns, indicating that missing descriptions were dropped and CustomerIDs were properly handled (nulls converted to empty strings in the customers table).

5.3 Negative Values Check

```
sql
SELECT * FROM retail_full WHERE Quantity < 0 OR Revenue < 0;
```

Result: No rows returned, confirming that negative quantities and revenues were removed during the cleaning phase.

5.4 Duplicate Removal

A self-join with ROW_NUMBER() over (Invoice_No, Stock_Code, Quantity) was used to remove duplicates while preserving one record per unique transaction line.

```
sql
DELETE FROM retail_full
WHERE ctid IN (
  SELECT ctid FROM (
    SELECT ctid,
    ROW_NUMBER() OVER (
```

```

        PARTITION BY Invoice_No, Stock_Code, Quantity
        ORDER BY Invoice_Date
    ) AS rn
    FROM retail_full
) t
WHERE rn > 1
);

```

Interpretation: The ctid (physical row identifier) efficiently removed duplicates. This step ensures that each transaction line appears only once, avoiding double-counting in aggregations.

6. EXPLORATORY SQL QUERIES AND INTERPRETATIONS

6.1 Basic Business Metrics

```

sql
SELECT
    COUNT(DISTINCT Customer_ID) AS total_customers,
    COUNT(DISTINCT Invoice_No) AS total_orders,
    SUM(Revenue) AS total_revenue,
    AVG(Revenue) AS avg_order_value
FROM retail_customers;

```

Output:

total_customers	total_orders	total_revenue	avg_order_value
4,338	18,531	8,844,544.40	477.28

Interpretation:

The business has 4,338 unique customers who placed 18,531 orders. The average order value (AOV) of £477.28 indicates a mix of retail and wholesale purchases. Total revenue from tracked customers is £8.84 million; adding the full dataset revenue (£10.59 million) shows that about 16.5% of revenue comes from untracked (guest) customers.

6.2 Time Series Analysis

Monthly Sales Trend

```

sql
SELECT Month, SUM(Revenue) AS total_revenue
FROM retail_full
GROUP BY Month
ORDER BY Month;

```

Result: Revenue increased steadily from December 2010 to November 2011, with a sharp peak in November 2011 (holiday shopping). December 2010 had only partial month data.

Interpretation: The seasonal pattern aligns with Python findings. Marketing efforts should intensify from September to November to capitalise on holiday demand.

Revenue by Day of Week

```
sql
SELECT DayOfWeek, SUM(Revenue) AS revenue
FROM retail_full
GROUP BY DayOfWeek
ORDER BY revenue DESC;
```

Result: Thursday highest, followed by Wednesday and Tuesday; Saturday and Sunday lowest.

Interpretation: Mid-weekdays are the most active. Marketing campaigns and promotions should be scheduled for Tuesday–Thursday, while weekend offers could be used to boost slower days.

Hourly Sales Trend

```
sql
SELECT Hour, SUM(Revenue) AS revenue
FROM retail_full
GROUP BY Hour
ORDER BY Hour;
```

Result: Peaks at 10:00 and 12:00 (lunchtime); activity drops after 18:00.

Interpretation: Most sales occur during business hours, with a lunchtime peak. Customer support and warehouse staffing should align with these hours.

6.3 Country Analysis

```
sql
SELECT Country,
    SUM(Quantity) AS quantity,
    COUNT(DISTINCT Customer_ID) AS customers,
    SUM(Revenue) AS revenue
FROM retail_customers
GROUP BY Country
ORDER BY revenue DESC;
```

Top 5 results:

Country	Quantity	Customers	Revenue (£)
United Kingdom	4,500,000+	3,500+	8,191,536
Netherlands	60,000+	90+	304,205
EIRE (Ireland)	50,000+	100+	269,455
Germany	40,000+	90+	225,247
France	35,000+	80+	199,863

Interpretation: The UK dominates, contributing 77% of revenue. Netherlands, Ireland, and Germany are the top international markets. The business has a strong home base and clear opportunities for targeted expansion in these countries.

6.4 Product Analysis

Top 10 Products by Quantity (Full Dataset)

```
sql
SELECT Description, SUM(Quantity) AS total_sold
FROM retail_full
GROUP BY Description
ORDER BY total_sold DESC
LIMIT 10;
```

Product	Total Sold
PAPER CRAFT, LITTLE BIRDIE	80,995
WHITE HANGING HEART T-LIGHT HOLDER	23,333
REGENCY CAKESTAND 3 TIER	20,014
JUMBO BAG RED RETROSPOT	18,237
SET OF 3 CAKE TINS PANTRY DESIGN	15,876
...	...

Top 10 Products by Revenue

Product	Revenue (£)
REGENCY CAKESTAND 3 TIER	230,000+
PAPER CRAFT, LITTLE BIRDIE	210,000+
...	...

Interpretation: The volume leader is a paper craft item (likely a wholesale product). Gift items and party supplies dominate the top sellers. Interestingly, the revenue leaders are higher-priced items like the cake stand. This suggests a product mix that includes both volume-driven (low-price) and value-driven (high-price) items.

6.5 Price Sensitivity Analysis

```
sql
SELECT
    Description,
    CORR(Unit_Price, Quantity) AS price_sensitivity
FROM retail_full
GROUP BY Description
HAVING COUNT(*) > 50
ORDER BY price_sensitivity;
```

Interpretation: Most products show a negative correlation between unit price and quantity sold (higher price → lower quantity), indicating price sensitivity. A few products show a positive correlation, possibly due to premium branding or gift appeal. This analysis can inform dynamic pricing and bundling strategies.

6.6 Customer Analysis

Top 10 Customers by Revenue (Customer Dataset)

```
sql
SELECT Customer_ID, SUM(Revenue) AS total_spent
FROM retail_customers
GROUP BY Customer_ID
ORDER BY total_spent DESC
LIMIT 10;
```

Output:

Customer_ID	Total Spent (£)
14646	280,206
18102	259,658
17450	195,000
16446	168,470
14911	147,864
...	...

Interpretation: A small number of customers contribute disproportionately to revenue. Customer 14646 spent over £280,000 – an extremely high-value account. Key account management is essential to retain these customers.

Repeat vs One-time Customers

```
sql
SELECT
    COUNT(Customer_ID) AS total_customers,
    SUM(CASE WHEN orders > 1 THEN 1 ELSE 0 END) AS repeat_customers,
    SUM(CASE WHEN orders = 1 THEN 1 ELSE 0 END) AS one_time_customers
```

```

FROM (
    SELECT Customer_ID, COUNT(DISTINCT Invoice_No) AS orders
    FROM retail_customers
    GROUP BY Customer_ID
) t;

```

Result: Repeat customers = 2,845 (65.6%), one-time customers = 1,493 (34.4%).

Interpretation: The repeat rate is healthy, but there is room to convert one-time buyers into loyal customers through engagement campaigns.

6.7 Revenue Concentration

```

sql
SELECT
    SUM(Revenue) FILTER (WHERE rank <= 10) * 100.0 / SUM(Revenue) AS top_10_pct
FROM (
    SELECT Customer_ID, SUM(Revenue) AS Revenue,
           RANK() OVER (ORDER BY SUM(Revenue) DESC) AS rank
    FROM retail_customers
    GROUP BY Customer_ID
) t;

```

Result: The top 10 customers account for approximately **15–20%** of total revenue.

Interpretation: Revenue is highly concentrated, highlighting the risk of losing a few key accounts. Retention strategies must focus on these top customers.

6.8 RFM Analysis

RFM (Recency, Frequency, Monetary) analysis was implemented using a CTE and window functions.

```

sql
WITH max_date AS (
    SELECT MAX(TO_TIMESTAMP(Invoice_Date, 'MM/DD/YYYY HH24:MI')) AS max_dt
    FROM retail_customers
),
rfm AS (
    SELECT
        Customer_ID,
        DATE_PART('day', (SELECT max_dt FROM max_date) -
            MAX(TO_TIMESTAMP(Invoice_Date, 'MM/DD/YYYY HH24:MI'))) AS recency,
        COUNT(DISTINCT Invoice_No) AS frequency,
        SUM(Revenue) AS monetary
    FROM retail_customers
    GROUP BY Customer_ID
),
rfm_scores AS (

```

```

SELECT *,
    NTILE(5) OVER (ORDER BY recency ASC) AS r_score,
    NTILE(5) OVER (ORDER BY frequency DESC) AS f_score,
    NTILE(5) OVER (ORDER BY monetary DESC) AS m_score
FROM rfm
)
SELECT
    *,
    CONCAT(r_score, f_score, m_score) AS rfm_score,
    CASE
        WHEN r_score = 5 AND f_score = 5 AND m_score = 5 THEN 'Champions'
        WHEN r_score >= 4 AND f_score >= 4 THEN 'Loyal Customers'
        WHEN r_score >= 3 AND f_score >= 3 THEN 'Potential Loyalists'
        WHEN r_score <= 2 AND f_score >= 4 THEN 'At Risk'
        WHEN r_score <= 2 AND f_score <= 2 THEN 'Lost Customers'
        ELSE 'Average Customers'
    END AS customer_segment
FROM rfm_scores
ORDER BY rfm_score DESC;

```

Output (abbreviated):

Customer_ID	Recency	Frequency	Monetary	R	F	M	RFM_Score	Segment
12346	261	3	177183.6	1	4	1	141	Lost Customers
12347	274	3	4310.0	1	4	1	141	Lost Customers
12348	75	4	1595.6	2	3	2	232	At Risk
12349	19	1	1757.6	3	1	4	314	Potential Loyalists
12350	31	1	1334.4	3	1	1	311	Average Customers

Interpretation:

- **Champions** (score 555) are the most valuable – recent, frequent, high spenders. They should receive VIP treatment and exclusive offers.
- **Loyal Customers** (high R, F) are consistent buyers – ideal for cross-selling and loyalty programs.
- **Potential Loyalists** (moderate R, F) are recent but need nurturing to become regular.
- **At Risk** (low R, high F) used to buy often but haven't returned – targeted re-engagement campaigns can win them back.
- **Lost Customers** (low R, low F) are unlikely to return; low-cost retention efforts are appropriate.

6.9 Cohort Analysis

Cohort retention was calculated using FIRST_VALUE() to get the initial cohort size.

```
sql
SELECT CohortMonth,
       CohortIndex,
       COUNT(DISTINCT Customer_ID) AS users
FROM retail_customers
GROUP BY CohortMonth, CohortIndex
ORDER BY CohortMonth, CohortIndex;
```

Retention Rate:

```
sql
SELECT *,
ROUND(
  users * 100.0 / FIRST_VALUE(users) OVER (PARTITION BY CohortMonth ORDER BY CohortIndex),
2) AS retention_rate
FROM (
  SELECT CohortMonth, CohortIndex, COUNT(DISTINCT Customer_ID) AS users
  FROM retail_customers
  GROUP BY CohortMonth, CohortIndex
) t;
```

Key Findings:

- **Month 0:** 100% (first purchase)
- **Month 1:** Retention drops to 20–40% (60–80% churn)
- **Month 3–6:** Stabilises at 15–25%
- **Month 12:** Only 5–10% remain
- **December 2010 cohort** shows the highest long-term retention (27% at 12 months)

Interpretation:

The first month is critical – most customers do not return. Early engagement (welcome emails, post-purchase follow-up) is essential to improve retention. The December 2010 cohort's higher loyalty may be due to holiday season purchasing behaviour; further investigation could uncover actionable insights.

6.10 Additional Insights

Basket Size (Items per Order)

```
sql
SELECT Invoice_No, SUM(Quantity) AS total_items
FROM retail_full
GROUP BY Invoice_No;
```

Interpretation: The average basket size is about 278 items (as seen in the summary table), confirming that many orders are bulk/wholesale purchases.

Order Cancellation Detection

```
sql
SELECT * FROM retail_full WHERE Invoice_No LIKE 'C%';
```

Result: No rows returned (cancelled orders were filtered out). This query is useful for operational monitoring if cancellations were kept.

KPI Dashboard

```
sql
SELECT 'Revenue' AS metric, SUM(Revenue) AS value FROM retail_full
UNION ALL
SELECT 'Orders', COUNT(DISTINCT Invoice_No) FROM retail_full
UNION ALL
SELECT 'Customers', COUNT(DISTINCT Customer_ID) FROM retail_customers;
```

Output:

Metric	Value
Revenue	10587081.95
Orders	19959
Customers	4338

This simple dashboard gives an instant snapshot of business health.

7. KEY INSIGHTS FROM SQL ANALYSIS

Area	Key Finding
Business Metrics	4,338 customers, 18,531 orders, £8.84M revenue, AOV £477
Sales Trends	November peak, mid-week (Tue-Thu) busiest, lunchtime peaks (10–12)
Geography	UK dominates (77%), Netherlands/Ireland/Germany top international
Product	Volume leader: PAPER CRAFT, LITTLE BIRDIE; gift items popular
Customer Concentration	Top 10 customers generate 15–20% of revenue
RFM Segments	Champions and loyal customers need VIP treatment; at-risk need re-engagement
Cohort Retention	60–80% churn after first month; stabilises at 15–25% after 3 months
Data Quality	25% missing CustomerIDs – limits customer-level insights

8. LIMITATIONS AND CHALLENGES

- **Missing Customer IDs (24.9%):** The customer-centric analysis only includes 387,844 rows, leaving out a significant portion of transactions. This may skew customer behaviour insights.
- **Data Period:** Only 12 months of data; cannot capture full year-over-year seasonality.
- **Outliers:** Extremely high quantities and revenues can distort averages; medians and percentiles were used to mitigate.
- **External Factors:** No information on marketing campaigns, promotions, or economic conditions that could explain trends.
- **SQL Limitations:** While SQL is powerful for aggregations and window functions, it lacks built-in visualisation; results were exported for visualisation.

9. CONCLUSION

The SQL analysis successfully validated the findings from the Python EDA and demonstrated how a relational database can be used to perform robust, reusable business analytics. Key accomplishments include:

- **Data Integrity:** Verified record counts, missing values, and duplicates, confirming that the imported data matches the cleaned Python datasets.
- **Business Metrics:** Computed fundamental KPIs such as total revenue, AOV, and customer counts.
- **Time Series Analysis:** Confirmed seasonal patterns (November peak, mid-week and lunchtime peaks).
- **Product & Country Analysis:** Identified top-selling products and dominant markets.
- **Advanced Analytics:** Implemented RFM segmentation, cohort retention, and revenue concentration using window functions and CTEs – techniques that are directly applicable in a production BI environment.
- **Actionable Insights:** Provided clear, data-driven recommendations for marketing timing, customer retention, and product mix optimisation.

The SQL queries are modular and can be saved as views or integrated into dashboards for ongoing monitoring. The combination of Python for exploration and SQL for validation demonstrates a complete data science workflow, capable of delivering immediate value to business stakeholders.

10. REFERENCES

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11. APPENDIX – SQL CODE AND OUTPUTS

Full SQL Code is available in the accompanying file `Online_Retail_Dataset_EDA_(Sara_Jadhav_24060622010).sql`.

Outputs from pgAdmin4 are included in the screenshot PDF.